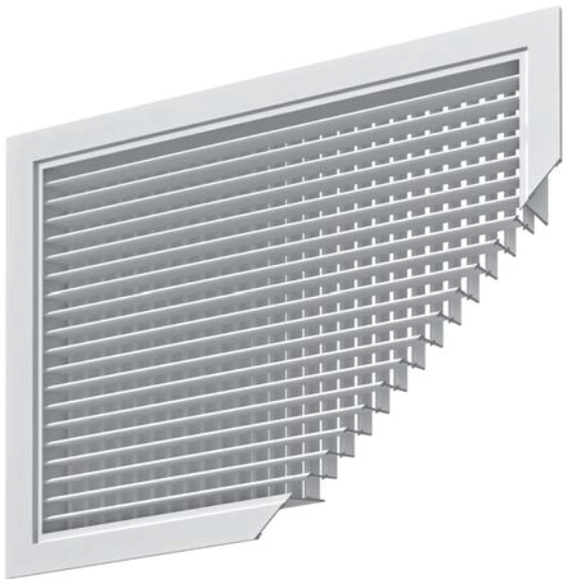
Double Deflection Grille DD/DD1



DD

Description
Aluminium ventilation grille equipped with individually adjustable blades that allow air flow direction. This can be provided with two rows of adjustable blades - called double deflection grille **DD** or a single row of adjustable blades - called **DD1**.

Application
Supply/ exhaust of the air flow

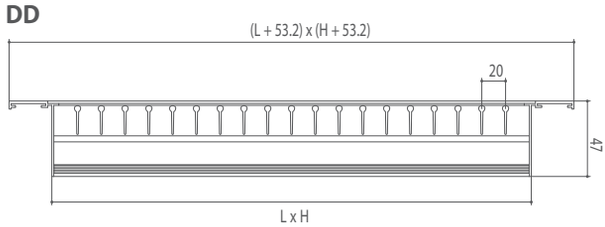
Characteristics
The front row blades are mounted perpendicular to the length **"L"**.
The grille is provided with mounting holes (standard).
A masked settlement frame **CC** can be provided (optional)
The mounting of the grille into the settlement frame is carried out using a clip mechanism to ease the mounting/ removal process.
Specially designed for paneled ceiling insertion (600 x 600 mm), the 595 x 595 mm (exterior) grille is provided without mounting holes.
The air flow can be adjusted by mounting an air control damper **OBD** on the grille inlet.
The grilles with length more than 500 mm are provided with intermediate shaft for reinforcement.

Accessories
Settlement frame made of galvanized sheet metal **CC**
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

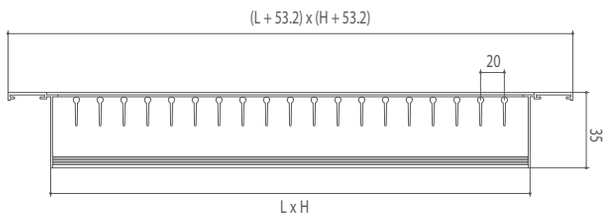
Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016.
Other RAL colors available on request.
Natural anodised aluminum.

Technical design

DD



DD1



Assembly Methods
Visible mounting holes on grille frame (standard).
Settlement frame (optional)

Selection table DD/DD1

Air flow (m^3/h)	LxH	200 x 100	300 x100	300 x 150	400 x 100	500 x 100	300 x 200	400 x 150	600 x 100	400 x 200	500 x 150	800x100	500x200	600 x 150	600 x200	800 x 150	500 x 300	800 x 200	600 x 300	800 x 300	1000 x 300
	Ak(m^2)	0.011	0.016	0.024	0.024	0.03	0.04	0.04	0.04	0.049	0.049	0.049	0.06	0.06	0.078	0.078	0.105	0.105	0.132	0.16	0.21
150	Veff[m/s]	3.8	2.6																		
	ΔPt [pa]	10.0	5.0																		
	X [m]	4.1	3.4																		
	NR [dB(A)]	-	-																		
200	Veffm/s]	5.0	3.5	2.3	2.3																
	ΔPt [pa]	18.0	8.0	4.0	4.0																
	X [m]	5.5	4.5	3.7	3.7																
	NR [dB(A)]	19.0	-	-	-																
300	Veff[m/s]	7.6	5.2	3.5	3.5	2.8	2.1	2.1	2.1												
	ΔPt [pa]	40.0	19.0	8.0	8.0	5.0	3.0	3.0	3.0												
	X [m]	8.2	6.8	5.6	5.6	5.0	4.3	4.3	4.3												
	NR [dB(A)]	28.0	21.0	-	-	-	-	-	-												
400	Veff[m/s]		6.9	4.6	4.6	3.7	2.8	2.8	2.8	2.3	2.3	2.3									
	ΔPt [pa]		3.4	15.0	15.0	10.0	5.0	5.0	5.0	4.0	4.0	4.0									
	X [m]		9.1	7.4	7.4	6.0	5.7	5.7	5.7	5.2	5.2	5.2									
	NR [dB(A)]		28.0	20.0	20.0	16.0	-	-	-	-	-	-									
500	Veff[m/s]			5.8	5.8	4.6	3.5	3.5	3.5	2.8	2.8	2.8	2.3	2.3							
	ΔPt [pa]			23.0	23.0	15.0	8.0	8.0	8.0	6.0	6.0	6.0	4.0	4.0							
	X [m]			9.3	9.3	8.3	7.2	7.2	7.2	6.5	6.5	6.5	5.9	5.9							
	NR [dB(A)]			25.0	25.0	21.0	-	-	-	-	-	-	-	-							
700	Veff[m/s]					6.5	4.9	4.9	4.9	4.0	4.0	4.0	3.2	3.2	2.5	2.5					
	ΔPt [pa]					29.0	17.0	17.0	17.0	11.0	11.0	11.0	7.0	7.0	4.0	4.0					
	X [m]					12.0	10.0	10.0	10.0	9.1	9.1	9.1	8.2	8.2	7.2	7.2					
	NR [dB(A)]					29.0	23.0	23.0	23.0	19.0	19.0	19.0	15.0	15.0	-	-					
900	Veff[m/s]						6.2	6.2	6.2	5.1	5.1	5.1	4.2	4.2	3.2	3.2	2.4	2.4			
	ΔPt [pa]						27.0	27.0	27.0	18.0	18.0	18.0	12.0	12.0	7.0	7.0	4.0	4.0			
	X [m]						13.0	13.0	13.0	12.0	12.0	12.0	11.0	11.0	9.2	9.2	8.0	8.0			
	NR [dB(A)]						29.0	29.0	29.0	25.0	25.0	25.0	21.0	21.0	16.0	16.0	-	-			
1200	Veff[m/s]									6.8	6.8	6.8	5.5	5.5	4.3	4.3	3.2	3.2	2.5	2.1	
	ΔPt [pa]									32.0	32.0	32.0	22.0	22.0	13.0	13.0	7.0	7.0	4.0	3.0	
	X [m]									16.0	16.0	16.0	14.0	14.0	12.0	12.0	11.0	11.0	9.5	8.6	
	NR [dB(A)]									32.0	32.0	32.0	22.0	22.0	23.0	23.0	17.0	17.0	-	-	
1500	Veff[m/s]												6.9	6.9	5.3	5.3	4.0	4.0	3.2	2.6	2.0
	ΔPt [pa]												34.0	34.0	20.0	20.0	11.0	11.0	7.0	5.0	3.0
	X [m]												18.0	18.0	15.0	15.0	13.0	13.0	12.0	11.0	9.4
	NR [dB(A)]												34.0	34.0	29.0	29.0	23.0	23.0	18.0	-	-

Annotation

- Ak [m²] - Free surface of the grille
- Veff[m/s] - Air velocity in grille
- Δ Pt [pa] - Pressure loss
- X [m] - Air jet length
- NR [dB(A)] - Noise level without the room attenuation
- Vk [m/s] - Speed at which airflow length was calculated "x" [m] - Vk [m/s] = 0.375 m/s

NR < 25 | 25≤NR≤35

Correction coefficients for other air velocities

Vk[m/s]		0.25	0.375	0.5
x[m]	A	1.5	1	0.75
	B	1.05	0.7	0.53

- A = with Coanda effect
- B = without Coanda effect

Order codes

Type + **OBD** + **CC** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: DD / DD-1
- **OBD**: Air control damper
- **CC**: Settlement frame
- **ADP IZ/N**: Insulated/ Non-insulated adapter
- **Dimensions**: **L X H**

L X H	L X H
200 x 100	500 x 300
300 x 100	600 x 100
300 x 150	600 x 150
300 x 200	600 x 200
400 x 100	600 x 300
400 x 150	800 x 100
400 x 200	800 x 150
500 x 100	800 x 200
500 x 150	800 x 300
500 x 200	1000 x 300

- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016; **R** Other RAL paints; Natural anodised